New rules, old habits: across seven team sports, each league keeps its own way of using the rules

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Abstract

Studies of a rule change usually report one average effect, for one league or several combined. We asked whether a rule change reorders leagues, in 101 leagues of seven sports. When the substitution limit rose from three to five, the leagues already ranked from most substitutions to fewest, as known for the biggest leagues. In their first season under five, the fourteen top leagues kept that order (rank correlation 0.89; one means the same order, zero unrelated orders). That is about as firmly as from one ordinary season to the next. A prediction for twelve second tiers, written before their records were read, held (0.73). They ranked like their country's top league. Across rule changes in other sports, leagues kept their order in four of eight tests. In every sport measured, a league stayed near its place among its sport's leagues from season to season. Under five, clubs inside a league differed more within a season than the leagues did, but their differences lasted less. Coaches who moved carried part of their old team's distance from its league. Their own part was about 4 substitutions a season. Rule-makers should judge a change league by league, against each league's own seasons.

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1 Introduction

Every team sport has rules on who may play and when. Football caps its substitutions, volleyball the players on its match sheet, handball its team time-outs and cricket each bowler's overs. Basketball and ice hockey let players come and go, and baseball now times its pitchers. These rules change often [1, 2]. Yet studies of a change usually report one average effect on results, running, injuries or the length of a game, for one league or several combined [3, 4, 5, 6]. Where one rule was studied in many countries, its effect differed between them [7, 8]. So an average over leagues can hide how each league responds. This paper asks what each league does with the same rule, in seven sports.

A rule can say what teams may do, and we call this the room it gives them. They choose what to do with that room. A speed limit is the everyday example. It sets how fast drivers may go.

Each road then settles at its own speed below the limit. In the same way, each league uses a rule's room in its own way. We call a league's own way of using the rules its habit. Statisticians call it a standardised league effect (Supplementary Section S6). A habit is measured from records the leagues already keep, such as substitutions per match or the number of players a team uses. The word describes only what teams do. A league's habit is the sum of its coaches' decisions. If habits differ, and keep differing, a rule-maker who reads the first season's average misjudges the leagues at both ends.

Football offers the cleanest case. In the summer of 2020 dozens of football leagues raised the substitution limit from three to five. IFAB, football's rule-maker, had allowed five at FIFA's request, to protect players' welfare during the pandemic [9, 10]. Each competition chose whether and when to use it [9]. So one new permission reached many leagues at dates of their own. In 2022 IFAB made this permission permanent [10, 11]. Every substitution is recorded with its minute, so what each league did with the new room can be counted.

Leagues are known to keep their own profiles, for example in home advantage and in fouls and cards [12, 13]. Analysts have shown that the biggest football leagues kept the same order in their use of substitutes from one season to the next [14, 15]. One of them called these differences lasting and explained them by each league's customs [14]. This paper tests whether leagues keep their order through a shared rule change, in football and in six other sports. We wrote most of these tests down before reading their data. We also wrote down nineteen predictions for seasons, leagues and competitions before reading their records.

We tested four claims. First, under five substitutions football's leagues would use the new room unevenly and keep their order. Second, leagues in other sports would also keep their order in how they use their rules. Third, a league's place among its sport's leagues would persist from season to season about as strongly in every sport. Fourth, a coach who moved would carry part of the old team's distance from its league's average. If these held, a rule change would be best judged league by league, against each league's own seasons.

2 Methods

2.1 Sports, leagues and records

Table 1 lists the sports in which these claims were tested, the rule change studied in each, and their leagues and records. Football's records come from transfermarkt-datasets, a public collection built from Transfermarkt, a football statistics website. We read its snapshot of 3 August 2026 [16]. The fourteen European leagues whose substitutions are counted hold 43,159 matches from 2015–16 to 2025–26. Other leagues enter some measures (Table 1).

Each league's limit in each season, three or five, was taken from the rules in force and checked against the substitutions made (Supplementary Table S1). A season in which the limit changed is split at the date of the change. We count substitutions per team-match, which is one team's side of one match. Supplementary Section S1 shows how the counts were checked against an independent

record, and corrected where the records fell short.

The records of basketball, volleyball, handball, ice hockey and baseball come from the leagues' and federations' own sites. Cricket's T20 leagues play matches of twenty overs a side. Their records come from Cricsheet, a public archive of ball-by-ball records [17].

2.2 Ethics

The study reused only public match records that already existed. These records were its only source on players and managers, so informed consent was outside its scope. Several tests read information related to health: the injury reasons that the records give for some substitutions, and players' absences from matchday squads. Both were counted by league, and only league values are reported. The records name professional players and managers, so they were treated as personal data, although they are public. Processing was for scientific research under the research provisions of the General Data Protection Regulation. Every published table and figure is aggregated to leagues, or to summaries across managers. The author treated these published records as archival data, which Trinity College Dublin's research ethics guidance exempts from approval [18]. So the study ran under that exemption. The same guidance grants approval only before a study begins.

2.3 What is measured

A rank correlation, or Spearman's, shows how well two orders agree: one for the same order, zero for unrelated orders and minus one for the reverse.

The spread between leagues in a season is the standard deviation of their means: roughly, a typical league's gap from their mean. A league's place on a measure is how many spreads its value sits above or below the mean of its sport's leagues that season. The persistence ϕ is the share of its place a league keeps from one season to the next. So after k seasons, a league keeps on average ϕ^k of its place. We call this slow loss of place drift. We measured one ϕ for each sport. We used the correlations between every pair of seasons, with the chance swings of single games taken out (Supplementary Section S6).

A team's distance from its league is its substitutions per team-match minus its league's mean that season. We average it over each coach's spell. Basketball's distance uses the five busiest players' share of minutes, and baseball's the players used beyond the starters. The coach's carry is the rank correlation between the old team's distance and the new team's, across the coaches who moved.

2.4 Pre-specified tests and statistics

Each test was written down as a short text, which we call a pre-specified test. It fixes what the test measures, its direction, which leagues qualify and what each outcome would be called. Each was saved, with its date, in the project's private online repository (Supplementary Section S4). Of

football's 26 pre-specified tests, thirteen were saved before their outcome was recorded. The other thirteen were written down only when they were run.

The other sports were tested on a split record. Each game went to one of two halves by a fixed rule that works like a coin toss. Half A was read freely, to choose each measure and pass mark. The tests were then saved before half B was read. The split guards against choosing a measure on the luck of single games. But both halves share the same leagues, so a measure chosen to suit those leagues could pass in both. The strongest evidence is therefore the seventeen predictions for new seasons, leagues or competitions. Each was written down before any of its records was read, and each was read once. One was rewritten before its records were read, and it is counted once.

A 95% interval gives the values that the data leave plausible (Supplementary Section S1). P is the chance that luck alone gives a result at least this strong. Most tests are one-sided, because each fixed its direction in advance. Such a test passes only on a link in its own direction. Every analysis added later is labelled stated. It came after the pre-specified tests, so it describes rather than tests [19].

Some values combine the leagues of several sports. They come from random-effects models, which give a combined value and the spread of the leagues around it [20]. They were fitted after the split's tests, so their values describe rather than test.

2.5 Sample size estimation and justification

The sources set the sample, so its size followed from them rather than from an estimate made in advance. Every league that the sources cover deeply enough to carry the measures was read. In football, fourteen of the source's thirty-one top leagues have seasons under three substitutions, so its coverage sets the sample (Supplementary Section S1). That is a justification by the whole available population [21, 22]. For such a sample, this journal asks for the smallest effect each test could detect [23]. The unit of each order test is the league, so its sample size is its number of leagues (Table 2).

For an order, the smallest effect of interest is a rank correlation of 0.5. Below it, the order in one season accounts for less than a quarter of the differences in a later one.

To detect a link, a test needs a 0.80 chance of passing at a one-sided P below 0.05. Twelve leagues detect only a true rank correlation of 0.69 or more. Fourteen leagues detect 0.65 or more. So the order and explanation tests could detect only links well above the smallest of interest. The largest test of coaches who moved detects 0.20 or more. These values come from exact counts up to nine leagues and from simulation beyond, in Python, numpy and SciPy (Supplementary Section S1).

3 Results

The tests open with a check. As expected, each rule moved what it was written to move (Table 2). The NHL, North America's top ice hockey league, brought in three-on-three overtime to settle

more tied games before a shootout [24]. Between December 2014 and 2016–17, five European leagues brought it in too. In them, three-on-three overtime multiplied the odds of settling a tied game before a shootout by 1.71. FIBA, basketball's world body, moved its three-point line out in 2010 [25]. The move cut each league's share of shots from three by about 9.9 per cent of that share. Baseball's pitch clocks were written to speed up the game [26]. They shortened games in all four leagues tested. The size of the line's and the clocks' effects differed between leagues (Supplementary Section S5). As Meyer and Klatt found, football's two extra substitutions moved playing time from starters to substitutes [27].

3.1 Football's leagues kept their order through the change

Under three substitutions football's leagues sat close to their limit (Figure 1). As earlier counts found, a team made about 2.85 substitutions a match [28, 29]. The leagues differed in how often teams used all three, from 76 per cent of team-matches in Scotland to 95 per cent in Portugal. So the leagues were already in an order, as analysts had seen in the biggest leagues [14, 15].

That order carried through the change. The fourteen leagues' order over 2015–16 to 2018–19 was much the same as in their first season under five (rank correlation 0.89). Under three, one season's order predicted the next season's at 0.80 to 0.87 (Supplementary Section S2). So the order held about as firmly as from one ordinary season to the next. These values were read after the change, so they describe rather than test.

A prediction then tested the order in twelve European second tiers. It was written before any of their records was collected. Their order under three predicted their order under five (rank correlation 0.73, one-sided $P = 0.0041$). In the seasons under five they also ranked much as their own country's top league did. That held for the eight whose top leagues we had read (rank correlation 0.90, one-sided $P = 0.0023$), and for all twelve (0.85; Supplementary Section S4). So the order is partly a country's, shared by its first and second tiers.

Under five, the top leagues spread apart in their substitutions, as expected when a limit that most teams reach is raised [30]. Had each league kept its share of the limit, the spread would have grown by five thirds. It grew more than that from 2020–21 to 2023–24, and about that much in the last two seasons (Figure 1b; Supplementary Section S2).

The change spread the leagues apart, and they kept their order. In the main pre-specified test, the order in the first season under five predicted the order in 2025–26 (rank correlation 0.73, one-sided $P = 0.0033$). On the eight leagues still unread when the test was written, the rank correlation was 0.57 ($P = 0.076$). The order from the seasons under three faded slowly. By 2025–26 its rank correlation with the new order was 0.60, as slow drift would give (Supplementary Section S2).

We tested four explanations of the order: money, rotation of the starting eleven, the value of the bench and absences from matchday squads. With thirteen or fourteen leagues, these tests could find only a strong link, and every link was weaker than that. Inside leagues, richer clubs used about as much of the limit as other clubs (mean rank correlation +0.02, 95% interval −0.16 to +0.20). In all, 51 tests read these and other explanations. After a correction for the number of tests, only

one passed, and it largely restates how many substitutions a league makes (Supplementary Section S3).

3.2 Leagues in other sports kept their order in four of eight tests

Across rule changes in other sports, the leagues kept their order in four of the eight pre-specified tests of the split record. Each test read few leagues. So only about four of the eight would pass if every order held at a rank correlation of 0.7 (Table 2). Across FIBA's longer three-point line of 2010, basketball's leagues kept their order in the share of shots from three. Across FIBA's shorter shot-clock reset of 2014, they kept it in the five busiest players' share of minutes and in the number of players given ten minutes. Volleyball's leagues kept theirs in the number of players used.

The other tests failed. Basketball's pace, a team's possessions per forty minutes, kept a weaker order. The 2014 change drew the leagues together in pace, as another pre-specified test expected. Volleyball's substitutions a set lost their order. Ice hockey's and baseball's tests read four leagues each, where only a perfect order can pass. Ice hockey's test read when a trailing team pulls its goalie, a choice its rules leave to the coach (Supplementary Section S4).

Predictions also read orders across seasons in these sports, and four of seven held. Baseball's eight leagues kept their ranks in pitchers used per team and game. The leagues lay far apart in pitchers used, so their prediction had expected this with near certainty. Cricket's thirteen T20 leagues kept their order in how many bowlers a side uses. The limit on each bowler's overs stayed the same, and some leagues changed their substitute rules (Tables 1 and 2). In basketball, two of four such predictions held, and ice hockey's just failed (Supplementary Section S6).

3.3 A league's place persists in every sport measured

As expected, in every sport measured, a league kept most of its place among its sport's leagues from one season to the next (Figure 2). Goals, points, volleyball's sets and cricket's runs kept 0.93 to 0.96 of their place a season, about as much. So leagues keep their character in general [12, 31], and their habit is part of it. The persistence of basketball, volleyball, ice hockey and baseball is consistent with one value, 0.969 a season. That value rests mostly on basketball's leagues (Figure 2a). Later seasons and new leagues, among them the NBA and the WNBA, gave values close to it, as their predictions expected (Table 2; Supplementary Section S6).

3.4 Football's clubs differed more than its leagues within a season

Under five, clubs inside a league differed more within one season than the leagues did. Their spread was about 0.36 substitutions a match, against 0.15 to 0.30 between leagues. But a club kept about 0.52 of its distance from its league into the next season, and a league about 0.91 of its place. These values allow for the chance swings of single matches (Supplementary Section S5).

3.5 Coaches carry part of a team's distance from its league

Coaches who moved carried part of their old team's distance from its league (Figure 3), as studies of managers' styles have found before [32, 33, 34]. Football managers carried their distance when they moved between clubs of one league, and when they changed league. Basketball's coaches carried it too, and a prediction on their moves since 2019 held. Baseball's two pre-specified tests, on its few moves, failed, so baseball is inconclusive. Combined over the three sports, the carry was a rank correlation of 0.30 (Supplementary Section S5).

The carry follows the coach rather than the club's choice of coach. In football, a coach's distance at the old club correlated only 0.08 with the new club's own distance in the season before. Allowing for that earlier distance, the carry was 0.26 in football and 0.31 in basketball. A model that separates managers from their clubs found a part that is the coach's own. It was small: about 0.13 substitutions a match, or 4 a season (Supplementary Section S5).

4 Discussion

Football's leagues kept their order through the change of limit, about as firmly as from one ordinary season to the next. The twelve second tiers read later kept theirs too. We read seventeen of the nineteen predictions written before their records were read, and twelve held. Of these passes, seven could plausibly have failed, among them the second tiers'. The other five would have passed under a much weaker claim too (Table 2; Supplementary Section S4).

These results fit how sport science sees a rule. A rule that gives room without saying how to use it is a task constraint [35]. Loosening a constraint widens the choices that work, and players and coaches settle among them [36, 37, 38]. In this view a competition organises itself [39]. Football shows this at the scale of whole leagues. The old limit of three held the leagues close to it. When teams are held at a limit, the count shows only that they wanted at least that many [30]. So raising the limit had to let the leagues spread apart. That part of the spread is the limit's doing. But the order beneath the limit was already there, and the limit did not set it.

Two readings fit the kept order. In one, each league raised its old level in proportion to the new limit. In the other, each league answered the change in its own way. The old limit held most teams at three, so the leagues' old levels understate how differently they wanted to use substitutes. Each league's level also drifts slowly, even when no rule changes. So its departure from the leagues' shared rise lasts from season to season (Supplementary Section S2). These counts therefore cannot tell the two readings apart. The claim here is the one both share: the leagues kept their order through the change.

Although clubs differ more within a season, a change is best judged by league. A rule is adopted competition by competition. The Premier League's clubs voted three times in 2020 against five substitutions, partly from concern that five would favour clubs with deeper squads [40, 41]. Under five, richer clubs used about as much of the limit as other clubs in their league. These records cannot say whether five favoured them. So the league is where a change is adopted, and where

differences in its use last longer.

4.1 Practical applications

A rule-maker weighing a change, from IFAB to a single competition, can forecast from the old rule which leagues will use it most. The forecast weakens over the seasons, as the old order fades (Supplementary Section S2). The gaps it forecasts are modest. In each season under five, the spread between football's leagues was 0.15 to 0.30 substitutions per team-match. Over a season of about 33 league matches, that is about 5 to 10 substitutions per club. Clubs within one league differed more, by about 12 substitutions a season, and a coach's own part was about 4 (Supplementary Section S5). A new limit holds teams back first where use is already near it. In 2025–26 the busiest league used 94.6 per cent of five, and the least busy 82.4 per cent.

So a change is best judged league by league. Methods for rules adopted at different times estimate an effect for each group of leagues that adopted it together [42]. Before the change, measure how each league uses the rule, from the records it already keeps. After it, compare each league with its own seasons before, not with the average of all leagues. Judge it over several seasons. After this change, use kept rising: every league made more substitutions in 2025–26 than in its first season under five. The leagues also took up the new room at different speeds (Supplementary Section S2). As reviews of rule changes advise, evaluate a change in several competitions [6], allow for their differences and register the plan publicly before the records arrive [2]. A study that combines leagues to estimate a rule's effect on injuries or running should give each league its own rise in substitutions, season by season. It should also report the effect league by league.

4.2 Relation to other work

The closest study uses the same rule change to study coaches' decisions [43]. It asks what the extra substitutions did within matches. This paper asks which leagues made them, and whether they kept doing so. A comparison of four of these leagues found that their results adapted alike to the change [5]. Here, their use of substitutions did not.

Several of these results first appeared in the first version of a preprint on nine sports [44]. This paper is that preprint's second version. The first version also claimed a law linking a league's load management to its timing. This version drops it: the two measures agree because of how they are built (Supplementary Section S7).

4.3 Limitations

No test here shows cause and effect. Every quantity describes how playing time was recorded, and a coach's reasons enter only where a team sheet records them. The records show how teams used their substitutions, not how often players were injured. No habit here is linked to results, running or injury. A test of the five-substitute rule's aim needs each league's injuries and hours of play, recorded under football's consensus methods [45].

The football records come from one public source, checked against an independent record only for some leagues and seasons (Supplementary Section S1). In 2020–21 and 2021–22 many records were incomplete. The rule change also came with a pandemic. But England, which kept three substitutions through the crowded season of 2020–21, did not use more of them (Supplementary Section S2).

Each league also sets rules of its own that were not measured in the other sports. Examples are roster sizes, limits on foreign players and MLB's limits on how many pitchers a team carries and how soon it may change them [46, 47]. These alone could keep a league in its place.

The tests are small. Most rest on four to 17 leagues, so they could detect only a strong link. The predictions on persistence would have failed only at a much weaker persistence, so their passes say little (Supplementary Section S6).

No public register holds the pre-specified tests. Their dates rest on the project's private repository, whose history was rewritten (Supplementary Section S4). The split record's tests were uploaded with their results, so only the order in which they were saved dates them.

When football's main order test was written down, the seasons under five of France, Germany, Italy and Spain had already been read. On the other eight leagues alone it fell short of its pass mark. It also measures persistence after the change, not across it (Supplementary Section S2). The second tiers share their country's order, so they are not an independent sample of leagues. In five of the eight countries one decision set both tiers' limit, so what carries the order is open (Supplementary Section S4).

Two predictions for a football rule of 2026–27 were not read, because this study ended before the end of that season (Table 2; Supplementary Section S4). The coach's carry rests on managers who moved, and estimates from few movers can be distorted [48]. Supplementary Section S5 discusses the two published criticisms of the design [49, 50].

5 Conclusion

A speed limit sets how fast drivers may go, and each road settles at its own speed below it. Football's substitution limit worked the same way. When it rose, each league settled at its own level below the new limit, in the order the leagues already had. A prediction written before twelve second tiers were read found that they kept their order too. In every sport measured, a league's place persisted from season to season. A rule sets the room, and each league uses that room in its own way. So a rule change should be judged league by league, against each league's own seasons before it, and over several seasons.

Data availability

All result tables, the analysis scripts and the test suite that rebuilds every number in this paper are in the archived record at DOI 10.5281/zenodo.24208722. The match records come from third-party

sources whose terms permit research use and not redistribution. The result tables are the minimum dataset needed to verify every claim. None of them names a player.

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Declaration of generative AI use

Anthropic's Claude, a generative AI assistant, was used under the author's direction to write and check the analysis scripts, search the literature and edit the text. The version used for this revision was Claude Opus 5.5, through the Claude Code tool. It was used because a single author could not otherwise build and check analyses of this size. The study design, the claims and their interpretation are the author's. Every reported quantity is computed from the committed data by the archived scripts and was verified by the author, who takes full responsibility for the content.

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Table 1: The seven sports, the rule change studied in each, and every record the analyses read. Every league here enters at least one reported test or estimate. The counts leave out the further football leagues that only the explanation models read (Supplementary Section S1). The two women's leagues of Supplementary Section S7 are left out too, because no table counts their records. Supplementary Tables S1 and S2 give each league's seasons. For football's first tiers, the number in brackets gives the fourteen European leagues whose substitutions are counted. Football's other first tiers enter the rises on their own windows or the timing measures (Supplementary Sections S2 and S7). Seasons named by one year are played within a calendar year. Cricket's bowling rule did not change. Supplementary Section S8 shows how each competition shares out its minutes.

Sport	Rule change read	Seasons	Leagues	Records
Football, first tiers	substitution limit raised from three to five, between 2019–20 and 2022–23 by league	2015–16 to 2025–26	21 (14)	47,937 matches
Football, second tiers	the same limit, in the second tiers of a prediction written before their records were read	2015–16 to 2025–26	12	33,485 matches
Basketball	three-point line moved out, 2010–11; shot clock reset after an offensive rebound shortened, 2014–15	2006–07 to 2025	22	55,051 games
Volleyball	more players allowed on the match sheet, at dates that differ by league	2006–07 to 2025–26	8	21,297 games
Handball	a third team time-out, an option each federation could take up; read in France, from January 2012	2011–12 to 2019–20	1	1,538 games
Ice hockey	tied games settled by three-on-three overtime before a shootout, at dates that differ by league	2011–12 to 2025–26	12	44,312 games
Baseball	a pitch clock, enforced from 2023 to 2025 by league	2008 to 2025	12	72,744 games
Cricket (T20)	none: at most a fifth of an innings per bowler throughout. Substitute rules changed in three leagues: the IPL's Impact Player from 2023, the ILT20's Super-Sub from 2023–24 and the BBL's X-Factor in 2020–21 and 2021–22	2019 to 2025	13	3,366 innings
Six other sports			68	198,308 games and innings
Seven sports			101	279,730 matches, games and innings

Table 2: Every pre-specified test of the paper's claims and checks, failures included, in three blocks. The predictions on new records were written down before any of their records was read. A prediction's chance to pass, if its claim held, is the clause's own figure, marked (clause). Otherwise it was computed after with the clause's simulator, from the fits as they stood when the clause was written, marked (after). Supplementary Section S4 gives each prediction's chance under a weaker rival. One prediction of the 2010 line was written on the day the line's first prediction failed, after that failure was recorded. Its spread was fitted on the competitions read before, among them those the first prediction read. The split record's tests were saved before half B was read. Supplementary Section S4 dates football's own tests, and Supplementary Section S7 describes its timing tests. The letter after each block's name begins the labels of its tests in the supplement. The smallest attainable P is the smallest P a test can reach with its number of leagues, or with the number of simulations used to compute it. A dash marks a test with no such floor. Power is a rank test's chance of passing when the true rank correlation is 0.7, with a one-sided P below 0.05 to pass. [a]The row includes a test that could hardly pass, or hardly fail, with its leagues.

Claim	Asked in	Leagues	Smallest attainable P	Power at 0.7	Chance to pass if the claim held	Held / asked
predictions on new records (G)						
One constant per rule (the 2010 line; three-on-three overtime)	basketball; ice hockey	1–2	–	–	0.73 to 0.91 (after)	1 of 2
One proportion of the share before, with a spread (the 2010 line)	basketball	6	–	–	0.95 (after)	1 of 1
One constant per second of clock (the first pitch clocks)	baseball	6	–	–	0.80 (clause)	0 of 1
A league keeps its place (basketball)	basketball	7	2.0×10^{-4}	0.49	0.48 to 0.50 (after)	2 of 4
The order holds over more leagues (ice hockey, baseball, cricket)[a]	ice hockey; baseball; cricket	6–13	2.5×10^{-5} to 0.0014	0.32 to 0.85	0.39 to 1.00 (clause)	2 of 3
One persistence a season across sports[a]	ice hockey; basketball; cricket	5–13	–	–	0.91 (after) to 1.00 (clause)	4 of 4
One carry across sports (the coach)	basketball	3	1.0×10^{-4}	1.00	0.69 (after)	1 of 1
Football's rules in new seasons and new leagues (written down for this version)[a]	football	12–14	1.0×10^{-5}	–	0.88 to 0.97 (clause)	1 of 1 (2 not read)
split record (R)						
A rule moves what it was written to move	basketball; ice hockey; baseball	4–6	–	–	–	3 of 3
The order of the leagues persists[a]	basketball; volleyball; ice hockey; baseball	4–8	2.5×10^{-5} to 0.042	0.22 to 0.60	–	4 of 8
A tightening draws the leagues together	basketball; baseball	4–7	5.0×10^{-4}	–	–	1 of 3
A loosened limit is used more, and lets the leagues differ	handball; volleyball; basketball; ice hockey	1–8	5.0×10^{-4}	–	–	4 of 6
A climb slows and levels off	handball; ice hockey; baseball	1–4	–	–	–	0 of 3
The habit before a change predicts the response	volleyball	5	–	–	–	0 of 1
The coach carries part of a team's distance from its league	basketball	3	1.0×10^{-4}	1.00	–	2 of 2
football's own pre-specified tests (C)						
The order persists across the five-substitution era (the order test)	football	12	2.1×10^{-9}	0.81	–	1 of 1
A material explanation accounts for the order (money, rotation, bench value, absences)	football	12–14	4.8×10^{-14} to 1.6×10^{-10}	0.85 to 0.92	–	0 of 4
A further explanation accounts for the order (culture; the league's own increment; two batteries)	football	13–31	1.2×10^{-34} to 1.6×10^{-10}	0.85 to 0.94	–	0 of 8
The coach's carry (the movers tests)	football; baseball	1–14	–	0.97 to 1.00	–	5 of 7
Timing before the change (the pre-change tests)[a]	football	8–12	2.5×10^{-5}	0.60	–	1 of 4
The timing slope on separate halves of the matches[a]	football	12	–	–	–	1 of 1
Resting between matches against timing within them[a]	football	8	2.5×10^{-5}	0.60	–	0 of 1

Figure captions

Figure 1. The fourteen football leagues carried the order they had under three substitutions through the change to five. (a) Substitutions per team-match in the fourteen European leagues, season by season, under the limit each season's regulations set. Each league is named by its country. The three highest leagues and the three lowest in 2025–26 are labelled, in the darker greys. The other eight are in light grey. England, dashed, used five in the restart of 2019–20 and went back to three for 2020–21 and 2021–22. The band marks 2019–20, which eight leagues split at the date of their change. Belgium's 2020–21 and Scotland's 2021–22 are split the same way, with a small grey box behind each part. Each part of a split season is drawn as its own point. Each mean is uncertain by about half a marker's height under three, and by about a marker's height under five (one standard error). Open markers are the fifteen league-seasons whose records were incomplete for more than a tenth of their team-matches (Supplementary Section S1). Several overlap in 2021–22. Ukraine's 2021–22 is not drawn, because almost none of its team-matches were kept. (b) The same counts as a share of each season's limit. On this scale, a league that kept its share of the limit stays level. (c) Each league's mean under three, over 2015–16 to 2018–19, against its mean under five, over every season under five. Each league keeps its marker from (a), and the leagues named in (a) are named here. The rank correlation ρ between the two means is printed with its 95% interval. It is description, read after the change. Open points are Denmark and England, which are outside the pre-specified test (Supplementary Section S2). The spread between the leagues depends on the scale. When the leagues share a limit, their order is the same on every scale (Supplementary Section S2).

Figure 2. In every sport measured, a league's place persists, about as firmly as its scoring. (a) The persistence ϕ, the share of its place a league keeps from one season to the next, with its 95% interval. Beside it are the leagues and pairs of seasons behind it, and each sport's weight in the pooled value, which combines the sports. Squares are sports, with their measures together, and circles are single measures. The open circle counts Japan's volleyball sets as played (Supplementary Section S4). The diamond is the pooled value over basketball, volleyball, ice hockey and baseball. Open squares are predictions, written down before their records were read, and they are not in the pool. Grey points are scoring, for comparison: goals, points and cricket's runs. The dashed line is the pooled value, and the dotted line a persistence of one. An interval that reaches past one comes from few seasons. The scale at the top gives the half-life under one rate: the number of seasons until a league would lose half of its place. (b) Basketball, which carries most of the pooled weight. Each point is the mean correlation of the leagues' places k seasons apart, with the chance swings of single games left in. Neighbouring values of k are grouped, and each point has its 95% interval over leagues. Each line is the mean correlation that a model predicts for the same pairs, with those swings left in. The solid line has one rate, basketball's ϕ. The dashed line has a permanent part, about 78 per cent of a league's place, plus a part that decays. It is fitted to the leagues' seasons, not to these means. It runs a little above the points, but it follows their levelling off, which one rate cannot.

Figure 3. Coaches carry part of a team's distance from its league, and every reading in three sports points the same way. Each row gives the rank correlation ρ between two distances from the league, with its 95% interval. One is the distance of the team a coach or manager left, and the other is the new team's. Beside it are the slope of a straight-line fit, with its interval, and the number of moves. Each sport has its own measure: substitutions a match in football, the five busiest players' share of minutes in basketball, and the players used beyond the starters in baseball. Black points, filled or open, enter the pooled value, which combines the sports, with their weight at right. The open point, the managers of MLB, the top baseball league of the United States and Canada, is inconclusive. Grey points are other readings of the same carry (Supplementary Section S5). The rows marked "given the destination" allow for the new club's own distance in the season before. Basketball's readings of every game and of half the games share most of their moves. The black diamond is the pooled value over the three sports, with its 95% interval, and the dashed line marks it. Neither ρ nor the slope is corrected for the chance swings of single games. Those swings blur any link, so both understate the carry.

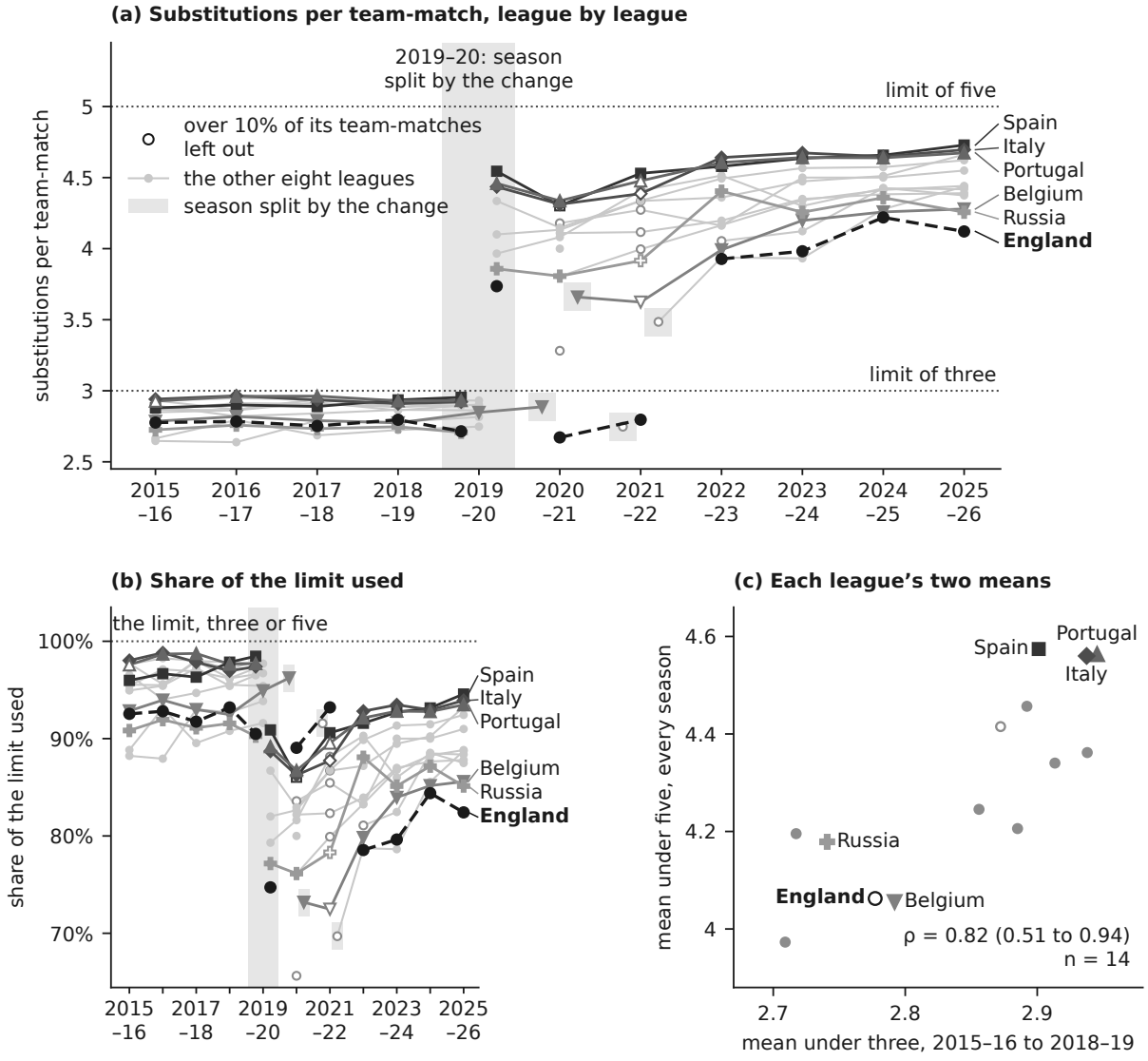

(a) Substitutions per team-match, league by league
2019–20: season split by the change
limit of five
limit of three
over 10% of its team-matches left out
the other eight leagues
season split by the change
substitutions per team-match
Spain
Italy
Portugal
Belgium
Russia
England
2015 –16
2016 –17
2017 –18
2018 –19
2019 –20
2020 –21
2021 –22
2022 –23
2023 –24
2024 –25
2025 –26
(b) Share of the limit used
the limit, three or five
share of the limit used
(c) Each league's two means
mean under five, every season
mean under three, 2015–16 to 2018–19
ρ = 0.82 (0.51 to 0.94)
n = 14

Figure 1

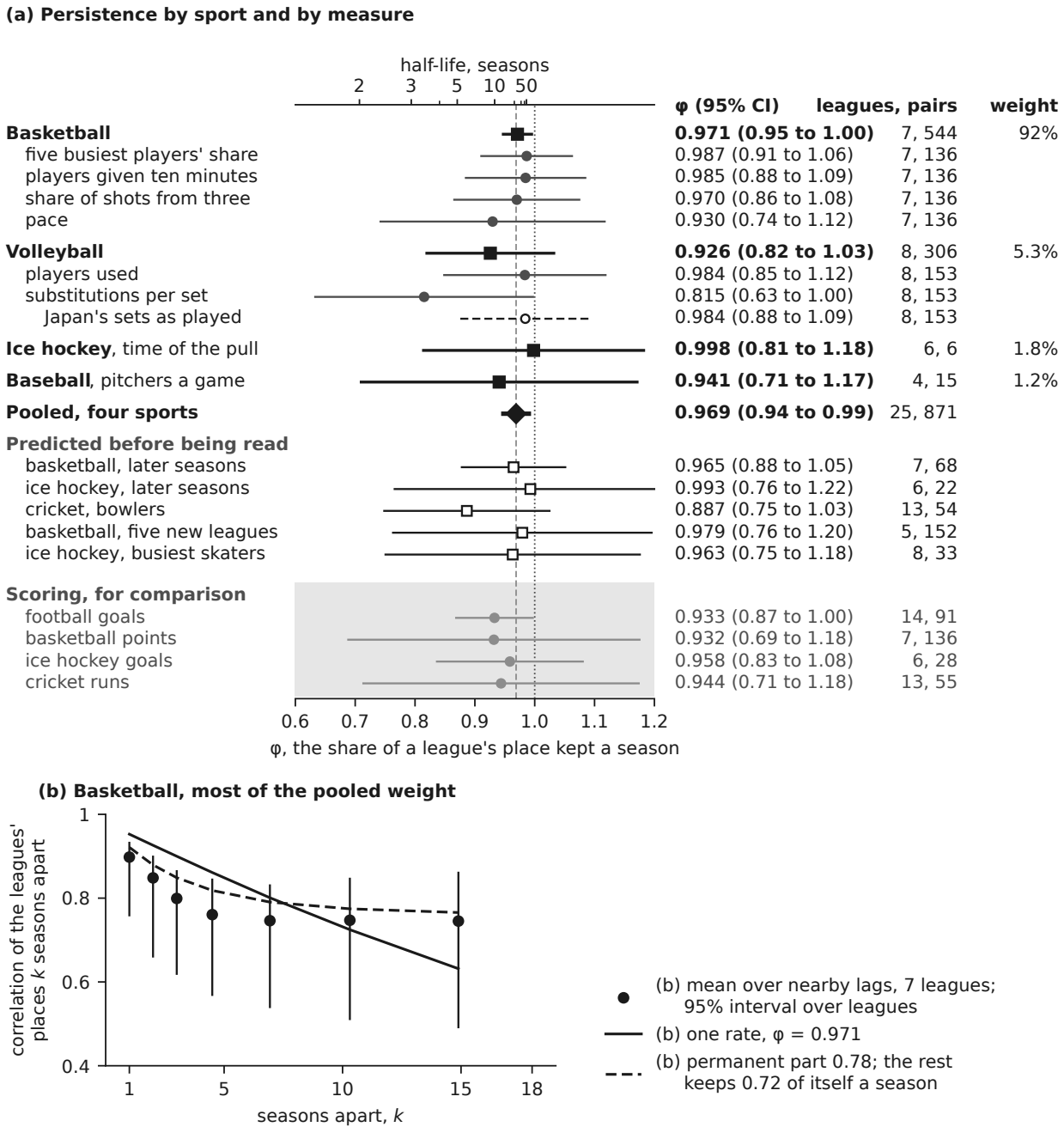

(a) Persistence by sport and by measure
half-life, seasons
2
3
5
10
50
φ (95% CI)
leagues, pairs
weight
Basketball
0.971 (0.95 to 1.00)
7, 544
92%
five busiest players' share
0.987 (0.91 to 1.06)
7, 136
players given ten minutes
0.985 (0.88 to 1.09)
7, 136
share of shots from three
0.970 (0.86 to 1.08)
7, 136
pace
0.930 (0.74 to 1.12)
7, 136
Volleyball
0.926 (0.82 to 1.03)
8, 306
5.3%
players used
0.984 (0.85 to 1.12)
8, 153
substitutions per set
0.815 (0.63 to 1.00)
8, 153
Japan's sets as played
0.984 (0.88 to 1.09)
8, 153
Ice hockey, time of the pull
0.998 (0.81 to 1.18)
6, 6
1.8%
Baseball, pitchers a game
0.941 (0.71 to 1.17)
4, 15
1.2%
Pooled, four sports
0.969 (0.94 to 0.99)
25, 871
Predicted before being read
basketball, later seasons
0.965 (0.88 to 1.05)
7, 68
ice hockey, later seasons
0.993 (0.76 to 1.22)
6, 22
cricket, bowlers
0.887 (0.75 to 1.03)
13, 54
basketball, five new leagues
0.979 (0.76 to 1.20)
5, 152
ice hockey, busiest skaters
0.963 (0.75 to 1.18)
8, 33
Scoring, for comparison
football goals
0.933 (0.87 to 1.00)
14, 91
basketball points
0.932 (0.69 to 1.18)
7, 136
ice hockey goals
0.958 (0.83 to 1.08)
6, 28
cricket runs
0.944 (0.71 to 1.18)
13, 55
0.6
0.7
0.8
0.9
1.0
1.1
1.2
φ, the share of a league's place kept a season
(b) Basketball, most of the pooled weight
correlation of the leagues' places k seasons apart
1
0.8
0.6
0.4
1
5
10
15
18
seasons apart, k
(b) mean over nearby lags, 7 leagues; 95% interval over leagues
(b) one rate, φ = 0.971
(b) permanent part 0.78; the rest keeps 0.72 of itself a season

Figure 2

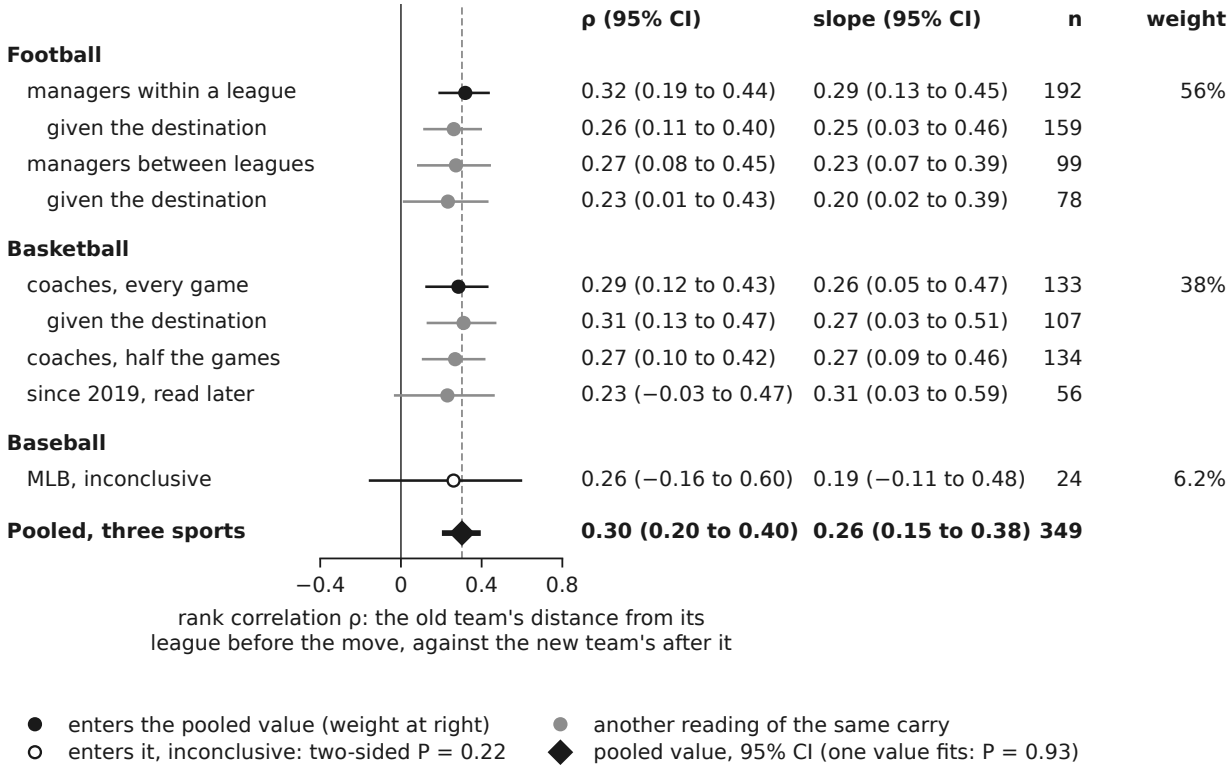

ρ (95% CI)
slope (95% CI)
n
weight
Football
managers within a league
0.32 (0.19 to 0.44)
0.29 (0.13 to 0.45)
192
56%
given the destination
0.26 (0.11 to 0.40)
0.25 (0.03 to 0.46)
159
managers between leagues
0.27 (0.08 to 0.45)
0.23 (0.07 to 0.39)
99
given the destination
0.23 (0.01 to 0.43)
0.20 (0.02 to 0.39)
78
Basketball
coaches, every game
0.29 (0.12 to 0.43)
0.26 (0.05 to 0.47)
133
38%
given the destination
0.31 (0.13 to 0.47)
0.27 (0.03 to 0.51)
107
coaches, half the games
0.27 (0.10 to 0.42)
0.27 (0.09 to 0.46)
134
since 2019, read later
0.23 (−0.03 to 0.47)
0.31 (0.03 to 0.59)
56
Baseball
MLB, inconclusive
0.26 (−0.16 to 0.60)
0.19 (−0.11 to 0.48)
24
6.2%
Pooled, three sports
0.30 (0.20 to 0.40)
0.26 (0.15 to 0.38)
349
−0.4
0
0.4
0.8
rank correlation ρ: the old team's distance from its league before the move, against the new team's after it
enters the pooled value (weight at right)
enters it, inconclusive: two-sided P = 0.22
another reading of the same carry
pooled value, 95% CI (one value fits: P = 0.93)

Figure 3